

Cdac Practical Exam

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Branch : Extc

Part A

Task 1

VM instances							
Filter Enter property name or value							
☐ Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	instance-20260610-060542	asia-south1-b			10.160.0.2 (nic0)	35.234.217.139 (nic0)	SSH ▾ ⋮

Task 2

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=...

SSH-in-browser

Linux instance-20260610-060542 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

akankshitwadte@instance-20260610-060542:~\$

+ Create Gemini API key + Create an agent

Task 3 :

The screenshot shows a web browser window with the address bar displaying the SSH URL: `ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...`. The browser tab is titled "SSH-in-browser". The terminal window displays the following content:

```
Linux instance-20260610-060542 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

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permitted by applicable law.
akankshitwadte@instance-20260610-060542:~$ mkdir practical_exam
akankshitwadte@instance-20260610-060542:~$ cd practical_exam
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir scripts
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir exams
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir logs
akankshitwadte@instance-20260610-060542:~/practical_exam$
```

Part B,C,D

The screenshot shows the same SSH-in-browser terminal window as above, but with additional commands and output. The terminal content is as follows:

```
Linux instance-20260610-060542 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

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individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
akankshitwadte@instance-20260610-060542:~$ mkdir practical_exam
akankshitwadte@instance-20260610-060542:~$ cd practical_exam
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir scripts
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir exams
akankshitwadte@instance-20260610-060542:~/practical_exam$ mkdir logs
akankshitwadte@instance-20260610-060542:~/practical_exam$ pwd
/home/akankshitwadte/practical_exam
akankshitwadte@instance-20260610-060542:~/practical_exam$ echo $USER
akankshitwadte
akankshitwadte@instance-20260610-060542:~/practical_exam$ echo hostname
hostname
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
!
akankshitwadte@instance-20260610-060542:~/practical_exam$ cd scripts
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ touch file1.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ touch file2.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ touch file3.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ touch file4.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ touch file5.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
!
akankshitwadte@instance-20260610-060542:~/practical_exam/scripts$ cd ..
akankshitwadte@instance-20260610-060542:~/practical_exam$ cp scripts/file1.txt outputs
akankshitwadte@instance-20260610-060542:~/practical_exam$ cp scripts/file2.txt outputs
akankshitwadte@instance-20260610-060542:~/practical_exam$ cd outputs
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ ls
file1.txt file2.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
!
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ mv file1.txt rename.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ ls
file2.txt rename.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
!
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ rm file2.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ ls
rename.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$
```

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=08

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-



SSH-in-browser



UPLOAD FILE



DOWNLOAD

```
akankshitwadte@instance-20260610-060542:~/practical_exam/outputs$ cd ..
akankshitwadte@instance-20260610-060542:~/practical_exam$ cd ..
akankshitwadte@instance-20260610-060542:~$ ls
practical_exam
akankshitwadte@instance-20260610-060542:~$ ls ./
practical_exam
akankshitwadte@instance-20260610-060542:~$
```

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=...



SSH-in-browser



UPLOAD FILE



DOWNLOAD FILE



```
akankshitwadte@instance-20260610-060542:~$ ls ./
practical_exam
akankshitwadte@instance-20260610-060542:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.3 168084 12512 ?        Ss   06:08   0:01 /sbin/init
root        2  0.0  0.0      0     0 ?        S    06:08   0:00 [kthreadd]
root        3  0.0  0.0      0     0 ?        I<   06:08   0:00 [rcu_gp]
root        4  0.0  0.0      0     0 ?        I<   06:08   0:00 [rcu_par_gp]
root        5  0.0  0.0      0     0 ?        I<   06:08   0:00 [slub_flushwq]
root        6  0.0  0.0      0     0 ?        I<   06:08   0:00 [netns]
root        8  0.0  0.0      0     0 ?        I<   06:08   0:00 [kworker/0:0H-events_highpri]
root       10  0.0  0.0      0     0 ?        I<   06:08   0:00 [mm_percpu_wq]
root       11  0.0  0.0      0     0 ?        I    06:08   0:00 [rcu_tasks_kthread]
root       12  0.0  0.0      0     0 ?        I    06:08   0:00 [rcu_tasks_rude_kthread]
root       13  0.0  0.0      0     0 ?        I    06:08   0:00 [rcu_tasks_trace_kthread]
root       14  0.0  0.0      0     0 ?        S    06:08   0:00 [ksoftirqd/0]
root       15  0.0  0.0      0     0 ?        I    06:08   0:00 [rcu_preempt]
root       16  0.0  0.0      0     0 ?        S    06:08   0:00 [migration/0]
root       18  0.0  0.0      0     0 ?        S    06:08   0:00 [cpuhp/0]
root       19  0.0  0.0      0     0 ?        S    06:08   0:00 [cpuhp/1]
root       20  0.0  0.0      0     0 ?        S    06:08   0:00 [migration/1]
root       21  0.0  0.0      0     0 ?        S    06:08   0:00 [ksoftirqd/1]
root       23  0.0  0.0      0     0 ?        I<   06:08   0:00 [kworker/1:0H-events_highpri]
root       26  0.0  0.0      0     0 ?        S    06:08   0:00 [kdevtmpfs]
root       27  0.0  0.0      0     0 ?        I<   06:08   0:00 [inet_frag_wq]
root       28  0.0  0.0      0     0 ?        S    06:08   0:00 [kauditd]
root       30  0.0  0.0      0     0 ?        S    06:08   0:00 [khungtaskd]
root       31  0.0  0.0      0     0 ?        S    06:08   0:00 [oom_reaper]
root       33  0.0  0.0      0     0 ?        I<   06:08   0:00 [writeback]
root       34  0.0  0.0      0     0 ?        S    06:08   0:00 [kcompactd0]
root       35  0.0  0.0      0     0 ?        SN   06:08   0:00 [ksmd]
root       36  0.0  0.0      0     0 ?        SN   06:08   0:00 [khugepaged]
root       37  0.0  0.0      0     0 ?        I<   06:08   0:00 [kintegrityd]
root       38  0.0  0.0      0     0 ?        I<   06:08   0:00 [kblockd]
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
SSH-in-browser
top - 06:30:52 up 22 min, 1 user, load average: 0.02, 0.03, 0.00
Tasks: 88 total, 1 running, 87 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.5 sy, 0.0 ni, 99.0 id, 0.2 wa, 0.0 hi, 0.0 si, 0.2 st
MiB Mem : 3924.7 total, 2497.1 free, 511.2 used, 1137.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3413.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1448 root        20   0 1999384 158048 107904 S   1.0   3.9   0:13.77 otelopscol
 1100 root        20   0 1278188 27500 15928 S   0.3   0.7   0:00.98 google_guest_ag
 1158 root        20   0 1276388 27396 14884 S   0.3   0.7   0:00.39 guest_telemetry
    1 root        20   0 168084 12512  9192 S   0.0   0.3   0:01.90 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
    6 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 netns
    8 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   10 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   11 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   12 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   13 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   14 root        20   0      0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/0
   15 root        20   0      0      0      0 I   0.0   0.0   0:00.28 rcu_preempt
   16 root        rt   0      0      0      0 S   0.0   0.0   0:00.00 migration/0
   18 root        20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   19 root        20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   20 root        rt   0      0      0      0 S   0.0   0.0   0:00.53 migration/1
   21 root        20   0      0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/1
   23 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   26 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   27 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   28 root        20   0      0      0      0 S   0.0   0.0   0:00.06 kauditd
   30 root        20   0      0      0      0 S   0.0   0.0   0:00.00 khungtaskd
   31 root        20   0      0      0      0 S   0.0   0.0   0:00.00 oom_reaper
```

```
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
SSH-in-browser
Linux instance-20260610-060542 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

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permitted by applicable law.
Last login: Wed Jun 10 06:09:43 2026 from 35.235.248.81
akankshitwadte@instance-20260610-060542:~$ cd practical_exam
akankshitwadte@instance-20260610-060542:~/practical_exam$ cd logs
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ touch secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chmod 640 secure_data.txt
chmod: cannot access 'secure_data.txt': No such file or directory
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ls
secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chmod +x
chmod: missing operand after '+x'
Try 'chmod --help' for more information.
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chmod +x secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chmod 640 secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ls
secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ls -l
total 0
-rw-r----- 1 akankshitwadte akankshitwadte 0 Jun 10 06:33 secure_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$

akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ echo $hostid secret_data.txt
secret_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ echo $userid secret_data.txt
secret_data.txt
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chown secret_data.txt
chown: missing operand after 'secret_data.txt'
Try 'chown --help' for more information.
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$
```

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...
ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?auth



SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE



```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ echo owner has permission r and w in 640
owner has permission r and w in 640
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ echo group has permission r in 640
group has permission r in 640
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ echo other have no permission in 640
other have no permission in 640
```

welcome

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ nano welcome.sh
```

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ./welcome.sh
```

welcome

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$
```

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...

ssh.cloud.google.com/v2/ssh/projects/quiet-light-498612-n0/zones/asia-south1-b/instances/instance-20260610-060542?authuser=0&hl=...



SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE



```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ chmod +x accept_user.sh
```

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ./accept_user.sh
```

enter username

akankshit

welcome akankshit

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$
```

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ nano largeroftwo.sh
```

```
akankshitwadte@instance-20260610-060542:~/practical_exam/logs$ ./largeroftwo.sh 5 6
```

./largeroftwo.sh: line 2: 5: command not found

6 is greater